The runcode package

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Abstract

runcode is a LaTeX package that executes programming source codes (including all command-line tools) from LaTeX and embeds the results in the resulting PDF file. Many programming languages can be used and any command-line executable can be invoked when preparing the PDF from a `.tex` file. runcode is also available on CTAN.

It is recommended to use this package in server mode together with the Python talk2stat package. Currently, the server mode supports Julia, MatLab, Python, and R. More languages will be added.

For more details, usage examples, and troubleshooting, refer to the package's GitHub repository at `https://github.com/Ossifragus/runcode`.

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1 Installation

You can simply put the `runcode.sty` file in your LaTeX project folder.

The server mode requires the talk2stat package. Some chunk commands (e.g., `\showChunk`) also require the advance-touch package. Install them from the command line with:

```
pip3 install talk2stat advance-touch
```

Note: `runcode` requires the `shell-escape` option when compiling a LaTeX document:

```
pdflatex --shell-escape yourfile.tex
```

When using AUCTeX or a similar IDE, you can avoid setting `shell-escape` manually by adding the following magic comment at the top of your `.tex` file:

```
% !TEX program = pdflatex --shell-escape
```

Replace `pdflatex` with `xelatex` or `lualatex` as appropriate.

2 Usage

2.1 Loading the package

```
\usepackage[options]{runcode}
```

Available options:

`cache` Use cached results (do not re-run code).

`fvextra` Use the fvextra package to display code.

`julia` Start a server for Julia (requires talk2stat).

`listings` Use the listings package to display code.

`matlab` Start a server for MatLab (requires talk2stat).

`minted` Use the minted package to display code (requires Pygments). This is the default.

`nominted` Use fvextra instead of minted (no syntax highlighting; no Pygments required).

`nohup` Use `nohup` when starting a server. Some editors (e.g. Emacs with AUCTeX) terminate child processes after compilation; this option keeps the server alive. **Must be declared before any language option**, e.g. `[nohup,R]` works but `[R,nohup]` does not. See Section 3 for tools that smooth the `nohup` workflow.

`python` Start a server for Python (requires talk2stat).

`R` Start a server for R (requires talk2stat).

reducedspace Reduce the vertical space around output boxes.

run Force source code to run (override cache).

stopserver Stop the server(s) when PDF compilation finishes.

2.2 Output box style

Output boxes are displayed using tcolorbox and can be customised with `\tcbset`, e.g.:

```
\tcbset{breakable, colback=red!5!white, colframe=red!75!black}
```

2.3 Code display style

The style of *source code* blocks (shown by `\showCode`, `\showChunk`, and the `\runLANGChunk` commands) depends on the package option used to display code:

- **minted** (default): use `\setminted` to control language-specific highlighting, e.g.:

  ```
  \setminted[r]{linenos, frame=single, bgcolor=bg, breaklines=true}
  ```

- **fvextra** or `nominted`: use `\fvset` to set options globally, e.g.:

  ```
  \fvset{frame=single, numbers=left, breaklines=true}
  ```

- **listings**: use `\lstset` to configure the style, e.g.:

  ```
  \lstset{basicstyle=\ttfamily\small, frame=single, numbers=left}
  ```

2.4 Basic commands

`\runExtCode` `\runExtCode{`⟨*prog*⟩`}{`⟨*src*⟩`}{`⟨*label*⟩`}[`⟨*flag*⟩`]` runs an external code file.

- ⟨*prog*⟩: the executable program.
- ⟨*src*⟩: the source file name.
- ⟨*label*⟩: the output file name (empty ⇒ uses counter `codeOutput`).
- `[`⟨*flag*⟩`]`: optional run control — omit or empty to use the global `runcode` Boolean; `run` to force execution; `cache` (or any other value) to use cached results.

Checksum-based cache invalidation (v2.5): when neither `cache` nor `run` is active, runcode computes an MD5 checksum of ⟨*src*⟩ and compares it with the checksum stored from the previous run. The script is re-executed only if the source has changed or if the output file is missing; otherwise the cached output is reused. This avoids unnecessary reruns without requiring the user to manage cache flags manually.

`\showCode` `\showCode{`⟨*lang*⟩`}{`⟨*src*⟩`}[`⟨*first*⟩`][`⟨*last*⟩`]` displays source code using minted, fvextra, or listings.

- ⟨*lang*⟩: programming language.
- ⟨*src*⟩: source file name.
- `[`⟨*first*⟩`]`/`[`⟨*last*⟩`]`: first and last line to display (optional; defaults to the whole file).

`\includeOutput` `\includeOutput{`⟨*label*⟩`}[`⟨*type*⟩`]` embeds the output of executed code.

- ⟨*label*⟩: output file name (must match ⟨*label*⟩ in `\runExtCode`; empty ⇒ uses counter).
- `[`⟨*type*⟩`]`: `vbox` (default) = verbatim in a box; `tex` = raw LaTeX; `inline` = inline text.

`\inln` `\inln{`⟨*prog*⟩`}{`⟨*code*⟩`}[`⟨*label*⟩`][`⟨*type*⟩`]` runs a short piece of code and displays the result inline.

- ⟨*prog*⟩: executable or language server.
- ⟨*code*⟩: source code.
- `[`⟨*label*⟩`]`: output file name (optional).
- `[`⟨*type*⟩`]`: controls display format and caching. Values: `inline` (default), `vbox`, `tex`; append `.cache` to any of these (e.g. `vbox.cache`) to reuse a cached result, re-running only if the output file is absent.

`\showChunk` `\showChunk{`⟨*lang*⟩`}{`⟨*src*⟩`}{`⟨*id*⟩`}[`⟨*begin*⟩`][`⟨*end*⟩`]` displays a labelled chunk of a source file. Chunks are delimited by comment lines containing `label===`⟨*id*⟩ (start) and `===end` (end) by default. These delimiters can be overridden with `[`⟨*begin*⟩`]` and `[`⟨*end*⟩`]`.

`\writeChunk` `\writeChunk{`⟨*lang*⟩`}{`⟨*src*⟩`}{`⟨*id*⟩`}[`⟨*begin*⟩`][`⟨*end*⟩`]` extracts the chunk identified by ⟨*id*⟩ from ⟨*src*⟩ and writes it to `generated/`⟨*src*⟩`-`⟨*id*⟩ on disk. This is called internally by `\showChunk` and `\runLANGChunk`, but can also be used standalone when only the extraction step is needed. In normal (non-cache) mode, extraction is skipped and the cached file reused whenever ⟨*src*⟩'s MD5 checksum matches the one recorded the last time it was extracted, so repeated `\showChunk` calls against an unchanged source are effectively free; editing ⟨*src*⟩ automatically invalidates every chunk cached from it, so no manual cache-clearing step is needed. In `cache` mode, the checksum check itself is skipped: an existing cached file is trusted outright with no subprocess call at all, matching `\runExtCode`'s and `\inln`'s behaviour in that mode (see Section 2).

2.5 Extended commands

`\runCodeIncOut` `\runCodeIncOut{`⟨*prog*⟩`}{`⟨*src*⟩`}[`⟨*flag*⟩`][`⟨*label*⟩`][`⟨*type*⟩`]` combines `\runExtCode` and `\includeOutput` in one call.

2.6 Language-specific shortcuts

Replace `LANG` with `Julia`, `MatLab`, `Python`, or `R` in the following commands.

`\runLANG` `\runLANG[`⟨*prog*⟩`]{`⟨*src*⟩`}{`⟨*label*⟩`}[`⟨*flag*⟩`]` runs an external `LANG` code file. `[`⟨*prog*⟩`]` is optional and defaults to the `talk2stat LANG` server.

\runLANGIncOut `\runLANGIncOut[⟨prog⟩]{⟨src⟩}[⟨flag⟩][⟨label⟩][⟨type⟩]` runs a `LANG` code file and embeds the output.

\inlnLANG `\inlnLANG[⟨prog⟩]{⟨code⟩}[⟨label⟩][⟨type⟩]` runs `LANG` source code and displays the result inline. If ⟨*code*⟩ is wrapped in triple backticks (`"‘code"‘`), it is sent directly to the server; otherwise it is written to a file first.

\runLANGChunk `\runLANGChunk[⟨prog⟩]{⟨src⟩}{⟨id⟩}[⟨flag⟩][⟨label⟩][⟨type⟩]` extracts the chunk ⟨*id*⟩ from ⟨*src*⟩, runs it, and embeds the output.

- `[⟨label⟩]`: output file name. When omitted, the label is derived automatically as ⟨*src*⟩`-`⟨*id*⟩ (e.g. `code/analysis.R-section2`).
- `[⟨type⟩]`: output display type (`vbox`, `tex`, `inline`, or their `.cache` variants).

For example:

```
\runR{code/analysis.R}{result1}
\runRIncOut{code/analysis.R}[][result1]
\runRChunk{code/analysis.R}{section2}
\inlnR{‘‘‘mean(c(1,2,3,4,5))‘‘‘}
```

2.6.1 Python batch mode

\runPythonBatch `\runPythonBatch[⟨src⟩][⟨label⟩]` runs a Python file in batch mode (no server). Requires the dill module (`pip3 install dill`), which provides session save/restore so that variables persist across calls.

3 Auxiliary tools

The following scripts are distributed with the package and address common workflow needs. Copy them into your project directory alongside your `.tex` file.

3.1 `wait_for_server.py`

Polls a talk2stat server until it is ready to accept connections. This eliminates the server-startup race condition that can occur in `nohup` mode, where the first `\runR` call may arrive before the server socket is open.

```
python3 wait_for_server.py [LANG [DIR [TIMEOUT]]]
```

- `LANG`: language server to wait for — `R`, `python`, `julia`, or `matlab` (default: `R`).
- `DIR`: working directory of the server (default: `./`).
- `TIMEOUT`: seconds before giving up (default: 120).

3.2 `runcode-Makefile.sample`

A sample `Makefile` for documents compiled in `nohup` server mode. Copy it to your project as `Makefile` and set the `MAIN` and `LANGS` variables. The `make all` target encodes three best practices:

1. Quit any existing server before starting a fresh one, so the new server picks up the current `.config` settings (including any updated `PIPETIMEOUT`).
2. Call `wait_for_rserver.py` to block until the server is ready before the first LaTeX pass.
3. After the draft pass (`--no-pdf` / `-draftmode`), send a language-appropriate no-op command as a sync barrier (`invisible(NULL)` for R, `pass` for Python, etc.) to ensure all queued scripts have finished writing their output files before the final passes begin.

PIPETIMEOUT note: the default timeout in the generated `.config` is 60 seconds. For long-running scripts, raise it by editing the `.config` file (e.g. `PIPETIMEOUT = 3600`) and restarting the server (`make stopserver`).

3.3 `consolidate.py`

Produces a self-contained copy of a runcode project in which all cached outputs are inlined into the `.tex` source, so the result compiles without talk2stat or any language runtime. This is useful for submission to Overleaf, a journal, or any recipient who does not have runcode installed.

```
python3 consolidate.py [--out DIR] [--engine CMD] [--no-compile]
                       [--exclude GLOB] MAIN.tex
```

The script copies the project to `standalone/` (configurable via `--out`), transforms all `.tex` files by replacing `\includeOutput`, `\inlnR`/`\inlnPython`/..., `\runRIncOut`, `\runRChunk`, etc. with the cached content from `generated/`, and substitutes `\usepackage{runcode}` with a minimal tcolorbox + listings shim. Run-only commands (`\runR`, `\runPython`, ...) are silently removed. A `make consolidate` target is provided in `runcode-Makefile.sample`.

4 Revisions

- **v2.6.2, August 2026:** The v2.6.1 MD5 checksum check fixed `\writeChunk`'s missing cache, but the checksum subprocess itself ran unconditionally — including under the `cache` package option, where it defeats the point: on Overleaf (which `make_overleaf.py`-style workflows put into `cache` mode) this meant one `python3` spawn per `\showChunk` call, on every compile, even with a warm cache. `\writeChunk` now checks `\ifruncode` first, matching `\runExtCode` and `\inln`: in `cache` mode it trusts an existing cached file outright with zero subprocess calls; the checksum check only runs in normal mode. Verified against the full factorial design (`\ifruncode` true/false × missing/fresh/stale cached file) in the new `examples/WriteChunkCacheTest/` regression test.
- **v2.6.1, August 2026:** `\writeChunk` re-scanned its whole source file, line by line, via a slow TeX-native loop on every call, with no caching — unlike every other command in the package. On documents with many `\showChunk` calls against large source files, across the multiple LaTeX passes a typical build requires, this dominated compile time (in one real-world document, a

single pass went from 3m33s to 11s once `\showChunk` calls were disabled). `\writeChunk` now applies the same MD5 checksum-based staleness check used elsewhere in the package (see the v2.6 entry below): the scan is skipped and the cached `generated/`⟨*src*⟩`-`⟨*id*⟩ file reused whenever ⟨*src*⟩ is unchanged since it was last extracted.

- **v2.6, June 2026:**
  1. Checksum-based cache invalidation: source files are fingerprinted with MD5; a script is re-executed only when its source has changed or when the output file is missing, avoiding unnecessary reruns.
  2. New `wait_for_rserver.py`: eliminates the server-startup race condition in `nohup` mode (see Section 3).
  3. New `runcode-Makefile.sample`: sample `Makefile` with server lifecycle management, startup synchronisation, and sync barrier (see Section 3).
  4. New `consolidate.py`: produces standalone LaTeX projects with all cached outputs inlined (see Section 3).
- **v2.4.1, June 10, 2026:** Bug fixes in `\writeChunk`: (1) replaced `mkdir -p` with Python `os.makedirs` for cross-platform subdirectory creation; (2) fixed stream exhaustion when more than 16 `\showChunk` calls appear in one document (streams are now pre-allocated at package load time); (3) removed erroneous `\write18{ad ...}` call in `\showChunk`. Added AUCTeX magic comment to example files.
- **v2.4, January 17, 2025:** (1) Put file names within `$$` for more robust warnings; (2) use the Python advance-touch package to create sub-folders for chunk-related commands.
- **v2.3, January 12, 2024:** Two bug fixes: (1) removed extra space after `\inlnX`; (2) fixed compilation error when an underscore appeared inside R code.
- **v2.2, September 8, 2023:** Added `\showChunk` basic command and `\runLANGChunk` commands for multiple languages.
- **v2.1, June 30, 2023:** Detokenize code passed to `\inln`. This prevents special LaTeX characters (e.g. backslash) from being escaped before reaching the interpreter. Thanks to kiryph for the report.
- **v2.0, June 23, 2023:** Added `\runCodeIncOut`; updated `\runExtCode` to override cache when output is missing; updated `\inln` to accept `cache` in `Arg4`. Thanks to kiryph for the suggestions.
- **v1.9, June 13, 2023:** Updated `\inln`: optional `Arg3` is the output file name, optional `Arg4` is the output type.
- **v1.8, January 18, 2023:** Added support for the listings package.
- **v1.7, August 20, 2022:** Renamed the `tmp/` folder to `generated/` (CTAN requirement); renamed the troubleshooting file.

- **v1.6, August 10, 2022:** Stop only configured/running servers; new `reducedspace` option; default server timeout changed to 60 s; expanded troubleshooting document.
- **v1.5, July 23, 2022:** Removed `utf8x` option from inputenc (conflict with hyperref).
- **v1.4, July 18, 2022:** Fixed a bug in cache mode.
- **v1.3, May 14, 2022:** Removed hard-coded minted options.
- **v1.2, May 3, 2022:** Added Python server and batch-mode options.
- **v1.1, April 17, 2021:** Added `nohup` option; improved error handling for missing code files and zero-byte outputs.

5 Contributing

We welcome contributions by opening issues or pull requests on GitHub. Additional example documents written using runcode are especially appreciated.

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